

Modeling the Impact of Dual Transmission, Vaccination, and Environmental Disinfection on Cholera Dynamics

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Abstract

This study investigates cholera dynamics using a modified deterministic SEIRB compartmental model that considers environmental exposure, direct human contact, vaccination, and environmental disinfection. The basis cholera model was modified by adding an exposed compartment to represent the incubation period before infected individuals become infectious. The human population is divided into susceptible S , exposed E , infectious I , and recovered or vaccinated R classes, while B represents the environmental concentration of *Vibrio cholerae*. The model was shown to be well-posed, and the disease-free and endemic equilibria were derived. The basic reproduction number R_0 was obtained using the next generation matrix method. Stability analysis shows that the disease-free equilibrium is locally asymptotically stable when $R_0 < 1$ and unstable when $R_0 > 1$, while the endemic equilibrium exists biologically when $R_0 > 1$. MATLAB simulations using `ode45` support the analytical results: stronger vaccination and disinfection drive the disease-related compartments toward zero, while weaker controls lead to disease persistence. The results suggest that reducing R_0 requires a combined public health strategy involving vaccination, sanitation, environmental disinfection, and reduced exposure to infectious individuals.

1 Introduction

Cholera is an acute *diarrhoeal* infection caused by eating or drinking food or water that is contaminated with the bacterium *Vibrio cholerae*. It is an extremely serious disease that can cause severe acute watery diarrhoea with severe dehydration. It takes between 12 hours and 5 days for a person to show symptoms after consuming contaminated food or water. Moreover, it affects both children and adults and can kill within hours if untreated [9, 10]. Despite the severity of the disease, most people infected with cholera are asymptomatic. However, the bacteria are present in their feces for about one to ten days after infection, revealing that the bacteria are shed back to the environment.

This phenomenon creates complex transmission routes on contracting the disease, and so, two of the pathways, direct household contact and environmental exposure, are considered in this study as both mechanisms operate simultaneously. Common outbreak control strategies include vaccination because it directly reduces susceptible individuals and environmental disinfection because it accelerates the decay rate of the said bacteria.

To achieve this, this framework modifies standard basis models of cholera, which commonly use compartmental structures for human and environmental transmission [2, 5, 1, 8], by introducing

an Exposed (E) compartment to capture the realistic biological delay between initial exposure and active infectivity.

Thus, the following control strategies are incorporated through a compartmental deterministic model to effectively analyze how these factors affect cholera transmission, considering the susceptible (S), exposed (E), infectious (I), recovered and vaccinated (R), and *Vibrio cholerae* environmental concentration (B) variables and through a non-constant population. Furthermore, this paper focuses on low bacterial concentration in the environment, and so the saturation parameter $\frac{B}{B+K}$ is simplified to $\frac{B}{K}$.

2 Model Formulation

The following assumptions are considered in the formulation of the model:

1. The total human population is divided into four compartments representing the model variables: the susceptible S , the exposed E , the infectious I , and the recovered or vaccinated (immune) R individuals. The variable B represents the concentration of the *Vibrio cholerae* bacteria in the environment.
2. The total human population is given by $N(t) = S(t) + E(t) + I(t) + R(t)$, while $B(t)$ is treated as an environmental variable and is not included in the human population.
3. Recruitment into the susceptible class occurs at a constant rate Λ . This replaces the immigration structure in the basis model of Onuorah et al. [6] and represents the general inflow of individuals into the susceptible population.
4. Cholera transmission occurs through two pathways. Susceptible individuals contract the disease indirectly from the environment at a rate $\omega\alpha_1 S \frac{B}{K}$, and through direct household contact with infectious individuals at a rate $\varphi\alpha_2 SI$.
5. Exposed individuals are infected but not yet infectious. They progress to the infectious class at a rate κ .
6. Susceptible individuals are vaccinated at a rate ν , which transitions them directly to the recovered or vaccinated (immune) compartment R .
7. Infectious individuals recover naturally at a constant rate γ , die due to cholera at a rate δ , and die naturally at a rate μ .
8. The environmental concentration of *Vibrio cholerae* increases due to shedding by infectious human hosts at a rate ϵI , and decreases due to natural decay at a rate τB and environmental disinfection at a rate ηB .
9. The environmental concentration of *Vibrio cholerae* is assumed to be relatively small compared with the scaling constant K , so the environmental term is simplified from $\frac{B}{K+B}$ to $\frac{B}{K}$.

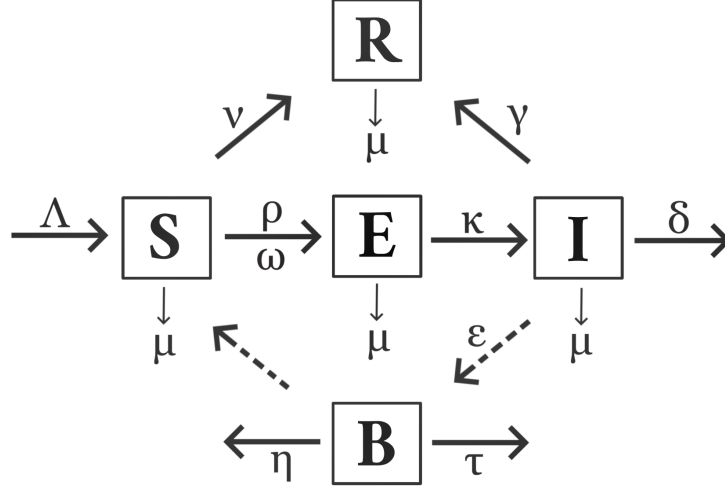


Figure 1: Schematic diagram of the modified SEIRB cholera transmission model.

Figure 1 shows the flow of individuals and bacteria in the modified cholera model. Susceptible individuals S may become exposed E through environmental contact with contaminated sources or direct contact with infectious individuals. Exposed individuals progress to the infectious class I , while infectious individuals may recover into R , die due to disease, or contribute to the environmental bacteria B . Vaccination moves susceptible individuals directly into the recovered or vaccinated class, while disinfection reduces the environmental bacteria concentration.

Parameter	Biological description	Value used	Unit	Source
a_0	Per-capita recruitment rate used to compute Λ	0.00913	day^{-1}	[4]
Λ	Recruitment rate into the susceptible population	9.19391	persons day^{-1}	computed from $a_0 N_{\text{ref}}$
μ	Natural death rate of the human population	5.48×10^{-5}	day^{-1}	[6]
φ	Effective direct household contact rate	0.0073	day^{-1}	[6]
ω	Effective environmental contact rate	0.00162	day^{-1}	[6]
α_1	Environmental infection coefficient	0.00015	dimensionless	[6]
α_2	Direct infection coefficient	0.00096	dimensionless	[6]
ν	Vaccination rate of susceptible individuals	varied	day^{-1}	[6]; varied in simulations
κ	Progression rate from exposed to infectious class	0.7143	day^{-1}	computed from [3]
γ	Recovery rate of infectious individuals	0.2	day^{-1}	[6]
δ	Disease-induced death rate	0.02	day^{-1}	[6]
ϵ	Bacterial shedding rate from infectious individuals	10	concentration day^{-1}	[6]
τ	Natural decay rate of bacteria	0.033	day^{-1}	[6]
η	Environmental disinfection rate	varied	day^{-1}	[6]; varied in simulations
K	Environmental bacterial scaling constant	500	concentration	[6]

Table 1: Model parameters, numerical values, units, and references. Parameter values applicable to the modified model were adapted from Onuorah et al. [6]. The recruitment rate Λ was computed using a_0 from Hartley et al. [4], while the progression rate κ was computed using the cholera incubation period reported by Azman et al. [3].

Table 1 summarizes the parameters used in the model. Most parameter values were adapted from the basis model of Onuorah et al. [6]. The recruitment rate Λ was computed from the per-capita recruitment rate a_0 , while the progression rate κ was introduced because the modified model includes an exposed compartment.

From these assumptions, the model is represented by the following system of nonlinear ordinary differential equations:

$$\begin{cases} \frac{dS}{dt} &= \Lambda - \omega\alpha_1 S \frac{B}{K} - \varphi\alpha_2 SI - (\nu + \mu)S, \\ \frac{dE}{dt} &= \omega\alpha_1 S \frac{B}{K} + \varphi\alpha_2 SI - (\kappa + \mu)E, \\ \frac{dI}{dt} &= \kappa E - (\gamma + \delta + \mu)I, \\ \frac{dR}{dt} &= \nu S + \gamma I - \mu R, \\ \frac{dB}{dt} &= \epsilon I - (\tau + \eta)B. \end{cases} \quad (1)$$

The first equation describes the change in the susceptible population. Susceptible individuals enter through recruitment, leave through environmental infection, direct contact infection, vaccination, and natural death. The second equation describes the exposed class, which increases through the two transmission pathways and decreases as exposed individuals progress to infection or die naturally. The third equation represents the infectious class, which increases from exposed individuals and decreases through recovery, disease-induced death, and natural death. The fourth equation describes the recovered or vaccinated class, while the fifth equation represents the environmental bacteria concentration, which increases through shedding by infectious individuals and decreases through natural decay and disinfection.

3 Mathematical Model Analysis

3.1 Well-Posedness of the Model

Let

$$X(t) = (S(t), E(t), I(t), R(t), B(t))^T.$$

Then the system can be written as $\frac{dX}{dt} = F(X)$, where $F(X)$ is the vector field determined by the right-hand side of (1). Since each component of F consists of constant, linear, and product terms in the state variables, F is continuous and continuously differentiable on $\mathbb{R}_+^5$.

The Jacobian matrix of the system is

$$J(X) = \begin{pmatrix} -\omega\alpha_1 \frac{B}{K} - \varphi\alpha_2 I - (\nu + \mu) & 0 & -\varphi\alpha_2 S & 0 & -\omega\alpha_1 \frac{S}{K} \\ \omega\alpha_1 \frac{B}{K} + \varphi\alpha_2 I & -(\kappa + \mu) & \varphi\alpha_2 S & 0 & \omega\alpha_1 \frac{S}{K} \\ 0 & \kappa & -(\gamma + \delta + \mu) & 0 & 0 \\ \nu & 0 & \gamma & -\mu & 0 \\ 0 & 0 & \epsilon & 0 & -(\tau + \eta) \end{pmatrix}.$$

Each entry of $J(X)$ is continuous. Hence, F is locally Lipschitz, and by the Picard–Lindelöf theorem, the model has a unique local solution for every nonnegative initial condition.

Define the feasible region

$$\Omega = \{(S, E, I, R, B) \in \mathbb{R}_+^5\}.$$

On the boundary of Ω , we have

$$\begin{aligned} \left. \frac{dS}{dt} \right|_{S=0} &= \Lambda > 0, \\ \left. \frac{dE}{dt} \right|_{E=0} &= \omega\alpha_1 S \frac{B}{K} + \varphi\alpha_2 SI \geq 0, \\ \left. \frac{dI}{dt} \right|_{I=0} &= \kappa E \geq 0, \quad \left. \frac{dR}{dt} \right|_{R=0} = \nu S + \gamma I \geq 0, \quad \left. \frac{dB}{dt} \right|_{B=0} = \epsilon I \geq 0. \end{aligned}$$

Thus, Ω is positively invariant.

Let $N(t) = S(t) + E(t) + I(t) + R(t)$. Adding the four human equations gives

$$\frac{dN}{dt} = \Lambda - \mu N - \delta I.$$

Since $I \geq 0$, it follows that $\frac{dN}{dt} \leq \Lambda - \mu N$. By comparison,

$$N(t) \leq \frac{\Lambda}{\mu} + \left(N(0) - \frac{\Lambda}{\mu} \right) e^{-\mu t},$$

and hence

$$N(t) \leq \max \left\{ N(0), \frac{\Lambda}{\mu} \right\}.$$

Moreover, since $\frac{dB}{dt} = \epsilon I - (\tau + \eta)B$ and $I \leq N$, the bacterial concentration $B(t)$ is also bounded. Therefore, all solutions remain nonnegative and bounded, and the model is well-posed in Ω .

3.2 Equilibrium Solutions

The next result concerns the existence of equilibrium points of the model. The equilibrium points of the model can be obtained by equating the reduced system to zero, that is,

$$\begin{cases} \Lambda - \omega\alpha_1 S \frac{B}{K} - \varphi\alpha_2 SI - (\nu + \mu)S = 0, \\ \omega\alpha_1 S \frac{B}{K} + \varphi\alpha_2 SI - (\kappa + \mu)E = 0, \\ \kappa E - (\gamma + \delta + \mu)I = 0, \\ \nu S + \gamma I - \mu R = 0, \\ \epsilon I - (\tau + \eta)B = 0. \end{cases}$$

3.2.1 The Disease Free Equilibrium $\mathcal{E}_0$

At the disease-free equilibrium, the disease is absent from the population. Hence, we set $I = 0$. From the infectious equation,

$$0 = \kappa E - (\gamma + \delta + \mu)I,$$

we obtain $E = 0$. From the bacterial equation,

$$0 = \epsilon I - (\tau + \eta)B,$$

we obtain $B = 0$. Therefore, $E = I = B = 0$.

Using the susceptible and recovered equations,

$$0 = \Lambda - (\nu + \mu)S, \quad 0 = \nu S - \mu R,$$

we obtain

$$S_0 = \frac{\Lambda}{\nu + \mu}, \quad R_{\text{DFE}} = \frac{\nu \Lambda}{\mu(\nu + \mu)}.$$

Thus, the disease-free equilibrium is

$$\mathcal{E}_0 = \left(\frac{\Lambda}{\nu + \mu}, 0, 0, \frac{\nu \Lambda}{\mu(\nu + \mu)}, 0 \right).$$

3.2.2 The Endemic Equilibrium $\mathcal{E}^*$

The endemic equilibrium represents the state where the cholera persists indefinitely in the community, maintaining a continuous chain of transmission.

Let:

$$\mathcal{E}^* = (S^*, E^*, I^*, R^*, B^*)$$

be an endemic equilibrium with all components positive. From the infectious and bacterial equations, we obtain

$$E^* = \frac{\gamma + \delta + \mu}{\kappa} I^*, \quad B^* = \frac{\epsilon}{\tau + \eta} I^*.$$

Define

$$Q = \varphi \alpha_2 + \frac{\omega \alpha_1 \epsilon}{K(\tau + \eta)}.$$

Substituting B^* into the exposed equation gives

$$0 = Q S^* I^* - (\kappa + \mu) E^*.$$

Using the expression for E^* and $I^* > 0$, we get

$$S^* = \frac{(\kappa + \mu)(\gamma + \delta + \mu)}{\kappa Q} = \frac{\Lambda}{(\nu + \mu) R_0}.$$

From the susceptible equation,

$$0 = \Lambda - QS^*I^* - (\nu + \mu)S^*.$$

Solving for I^* gives

$$I^* = \frac{\kappa\Lambda(R_0 - 1)}{(\kappa + \mu)(\gamma + \delta + \mu)R_0}.$$

Therefore,

$$E^* = \frac{\Lambda(R_0 - 1)}{(\kappa + \mu)R_0},$$

$$B^* = \frac{\epsilon\kappa\Lambda(R_0 - 1)}{(\tau + \eta)(\kappa + \mu)(\gamma + \delta + \mu)R_0}.$$

Finally, from the recovered equation,

$$R^* = \frac{\nu S^* + \gamma I^*}{\mu}.$$

Thus, $\mathcal{E}^* = (S^*, E^*, I^*, R^*, B^*)$. The endemic equilibrium exists biologically when $R_0 > 1$.

3.3 Basic Reproduction Number

The basic reproduction number, R_0 , is a crucial epidemiological metric that quantifies the expected number of secondary infections generated by a single infected individual in a completely susceptible population. We compute R_0 using the next generation matrix method. The infected variables are $x = (E, I, B)$. The new infection terms $\mathcal{F}$ and transition terms $\mathcal{V}$ are

$$\mathcal{F} = \begin{pmatrix} \omega\alpha_1 S \frac{B}{K} + \varphi\alpha_2 SI \\ 0 \\ 0 \end{pmatrix}, \quad \mathcal{V} = \begin{pmatrix} (\kappa + \mu)E \\ (\gamma + \delta + \mu)I - \kappa E \\ (\tau + \eta)B - \epsilon I \end{pmatrix}.$$

At the disease-free equilibrium, $S_0 = \frac{\Lambda}{\nu + \mu}$. The corresponding Jacobian matrices with respect to (E, I, B) are

$$F = \begin{pmatrix} 0 & \varphi\alpha_2 S_0 & \frac{\omega\alpha_1 S_0}{K} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} \kappa + \mu & 0 & 0 \\ -\kappa & \gamma + \delta + \mu & 0 \\ 0 & -\epsilon & \tau + \eta \end{pmatrix}.$$

Since $R_0 = \rho(FV^{-1})$, we obtain

$$R_0 = \frac{\kappa\Lambda}{(\nu + \mu)(\kappa + \mu)(\gamma + \delta + \mu)} \left(\varphi\alpha_2 + \frac{\omega\alpha_1\epsilon}{K(\tau + \eta)} \right).$$

3.4 Stability Analysis

Theorem 1. *The disease-free equilibrium $\mathcal{E}_0$ is locally asymptotically stable if $R_0 < 1$, and unstable if $R_0 > 1$.*

Proof. At $\mathcal{E}_0$, the Jacobian matrix has two negative eigenvalues,

$$\lambda_1 = -\mu, \quad \lambda_2 = -(\nu + \mu).$$

The remaining eigenvalues are determined by the infected subsystem matrix

$$M_0 = \begin{pmatrix} -(\kappa + \mu) & \varphi\alpha_2 S_0 & \omega\alpha_1 \frac{S_0}{K} \\ \kappa & -(\gamma + \delta + \mu) & 0 \\ 0 & \epsilon & -(\tau + \eta) \end{pmatrix}.$$

Let $a = \kappa + \mu$, $c = \gamma + \delta + \mu$, $d = \tau + \eta$, $p = \varphi\alpha_2 S_0$, and $q = \omega\alpha_1 S_0/K$. The characteristic equation of M_0 is

$$(\lambda + a)(\lambda + c)(\lambda + d) - \kappa p(\lambda + d) - \kappa \epsilon q = 0.$$

Equivalently,

$$\lambda^3 + A_1 \lambda^2 + A_2 \lambda + A_3 = 0,$$

where

$$A_1 = a + c + d, \quad A_2 = ac + ad + cd - \kappa p,$$

and

$$A_3 = acd - \kappa p d - \kappa \epsilon q = acd(1 - R_0).$$

By the Routh–Hurwitz criterion, all roots have negative real parts if

$$A_1 > 0, \quad A_2 > 0, \quad A_3 > 0, \quad A_1 A_2 > A_3.$$

Since $a, c, d > 0$, we have $A_1 > 0$. If $R_0 < 1$, then $A_3 > 0$. Moreover, $R_0 < 1$ implies

$$\kappa p + \frac{\kappa \epsilon q}{d} < ac,$$

and so

$$A_2 = ac + ad + cd - \kappa p > ad + cd > 0.$$

Hence, $\mathcal{E}_0$ is locally asymptotically stable when $R_0 < 1$.

If $R_0 > 1$, then $A_3 < 0$. Hence, the characteristic polynomial has at least one root with positive real part, and $\mathcal{E}_0$ is unstable. $\square$

Theorem 2. *Suppose $R_0 > 1$. The endemic equilibrium $\mathcal{E}^*$ is locally asymptotically stable whenever the Routh–Hurwitz conditions for the corresponding fourth-degree characteristic polynomial are satisfied.*

Proof. At $\mathcal{E}^*$, one eigenvalue is $-\mu < 0$. The remaining eigenvalues are determined by the reduced matrix involving S, E, I , and B :

$$M^* = \begin{pmatrix} -(m+h) & 0 & -p^* & -q^* \\ h & -a & p^* & q^* \\ 0 & \kappa & -c & 0 \\ 0 & 0 & \epsilon & -d \end{pmatrix},$$

where

$$a = \kappa + \mu, \quad c = \gamma + \delta + \mu, \quad d = \tau + \eta, \quad m = \nu + \mu, \\ h = \omega\alpha_1 \frac{B^*}{K} + \varphi\alpha_2 I^*, \quad p^* = \varphi\alpha_2 S^*, \quad q^* = \omega\alpha_1 \frac{S^*}{K}.$$

Let the characteristic polynomial of M^* be

$$\lambda^4 + b_1\lambda^3 + b_2\lambda^2 + b_3\lambda + b_4 = 0.$$

By the Routh–Hurwitz criterion for a fourth-degree polynomial, all roots have negative real parts if

$$b_1 > 0, \quad b_2 > 0, \quad b_3 > 0, \quad b_4 > 0, \\ b_1b_2 > b_3, \quad b_1b_2b_3 > b_3^2 + b_1^2b_4.$$

Therefore, under these conditions, the endemic equilibrium is locally asymptotically stable.

In summary, the stability for both disease-free and endemic equilibrium rely heavily on R_0 . If $R_0 > 1$, then there is a unique strictly positive solution for I^* (and consequently positive solutions for E^* and B^*), and thus there is a unique endemic equilibrium. If $R_0 \leq 1$, the biologically feasible positive equilibrium does not exist, leaving the disease free equilibrium as the only state. $\square$

4 Numerical Simulations

Numerical simulations were performed using MATLAB with the `ode45` solver [7] to support the analytical results obtained above. The simulations use the same model in (1). Since the immigration term in the reference model was removed, the recruitment term was computed as

$$\Lambda = a_0 N_{\text{ref}}.$$

The per-capita recruitment rate $a_0 = 0.00913$ was taken from Hartley et al. [4]. For the reference population, we use

$$N_{\text{ref}} = 1000 + 5 + 2 + 0 = 1007.$$

Thus, we obtain

$$\Lambda = 0.00913(1007) = 9.19391.$$

The progression rate from exposed to infectious individuals was computed using

$$\kappa = \frac{1}{T_E},$$

where T_E represents the average exposed period. Since Azman et al. estimated the median incubation period of toxigenic cholera to be 1.4 days [3], we set

$$\kappa = \frac{1}{1.4} = 0.7143.$$

Two cases were considered to illustrate the threshold behavior of the model. The same initial conditions were used for both the disease-free and endemic cases so that the difference in behavior is due to the control parameters and not to different starting populations. The initial conditions are summarized in Table 2.

Initial Condition	Disease-Free Case	Endemic Case
$S(0)$	1000	1000
$E(0)$	5	5
$I(0)$	2	2
$R(0)$	0	0
$B(0)$	20	20

Table 2: Initial conditions used in the numerical simulations.

The disease-free case uses stronger vaccination and environmental disinfection, while the endemic case uses weaker vaccination and disinfection. The corresponding threshold results are given in Table 3.

Case	ν	η	Threshold Result
Disease-free case	0.02	4	$R_0 \approx 0.0146 < 1$
Endemic case	0.0001	0.01	$R_0 \approx 1.92 > 1$

Table 3: Simulation cases used to illustrate the threshold behavior of the model. The parameters ν and η were varied from the control parameters in Onuorah et al. [6] to produce disease-free and endemic scenarios.

In all MATLAB time-series plots, the same color convention is used for consistency: susceptible S is blue, exposed E is orange, infectious I is red, recovered or vaccinated R is green, and environmental bacteria B is magenta.

For the disease-free case, the numerical solution confirms that the disease-related variables approach zero. The human compartment plot shows the overall behavior of the population, while the separate plots for $E(t)$, $I(t)$, and $B(t)$ make the disease-related variables more visible. These variables are small compared with the susceptible and recovered populations, so they are difficult to distinguish when all compartments are plotted together.

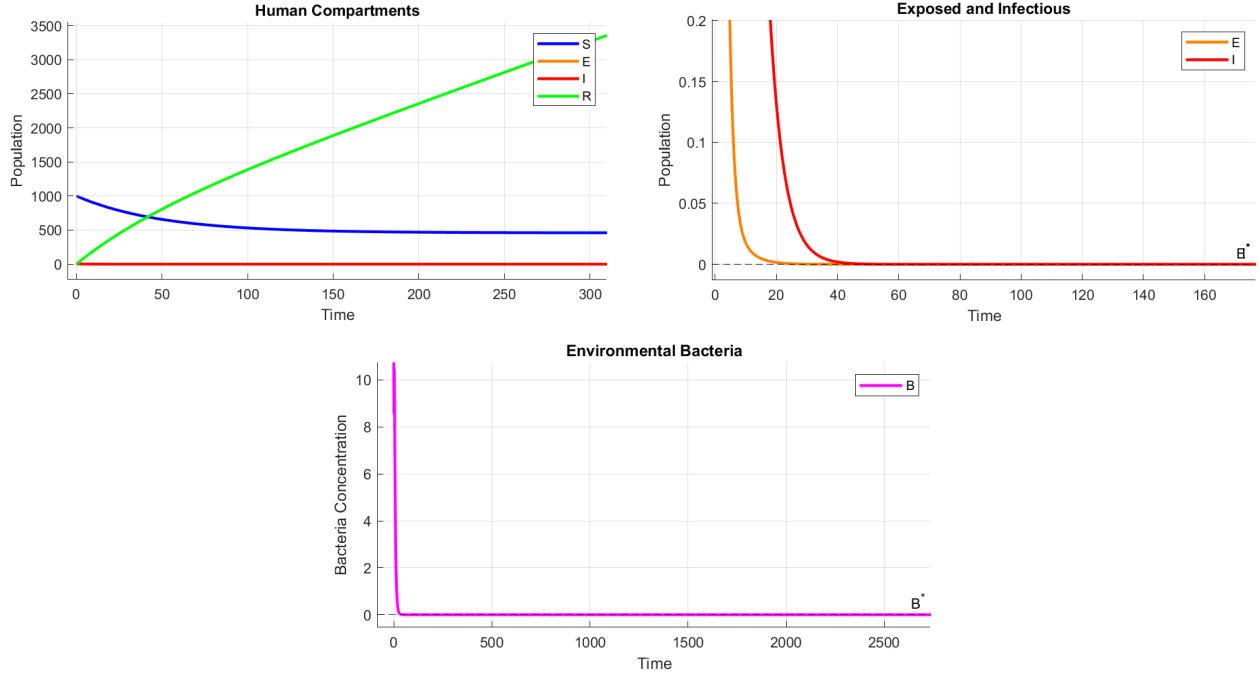


Figure 2: Disease-free simulation results showing the human compartments, exposed and infectious classes, and environmental bacteria.

Figure 2 shows that $E(t)$, $I(t)$, and $B(t)$ decrease toward zero when $R_0 < 1$. This supports the analytical result that the disease-free equilibrium is locally asymptotically stable under stronger vaccination and disinfection.

For the endemic case, the numerical solution shows disease persistence. The exposed, infectious, and bacterial compartments do not vanish, but instead approach positive values. This agrees with the analytical result that the endemic equilibrium exists biologically when $R_0 > 1$. A longer time interval is used because the natural death rate μ is very small, causing some compartments, especially $R(t)$, to approach equilibrium more slowly.

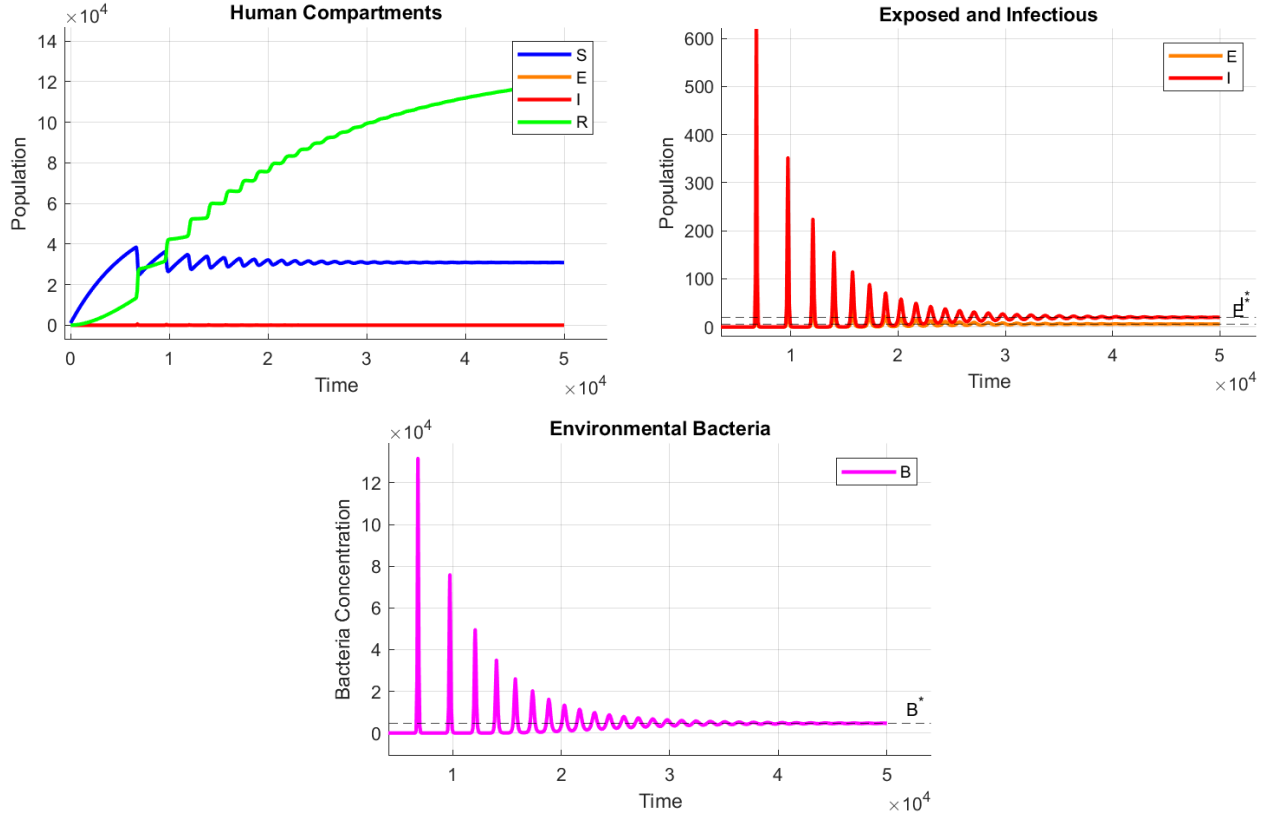


Figure 3: Endemic simulation results showing the human compartments, exposed and infectious classes, and environmental bacteria.

Figure 3 shows that the disease-related compartments persist when $R_0 > 1$. The separate plots for $E(t)$, $I(t)$, and $B(t)$ are included because these variables are much smaller in scale than the main human population compartments and are therefore not clearly visible in the combined human compartment plot.

The stability simulations further support the analytical stability results. Several initial conditions were tested to observe whether the solutions move toward the predicted equilibrium points.

Figure 3: Stability of DFE, $R_0 = 0.0146 < 1$

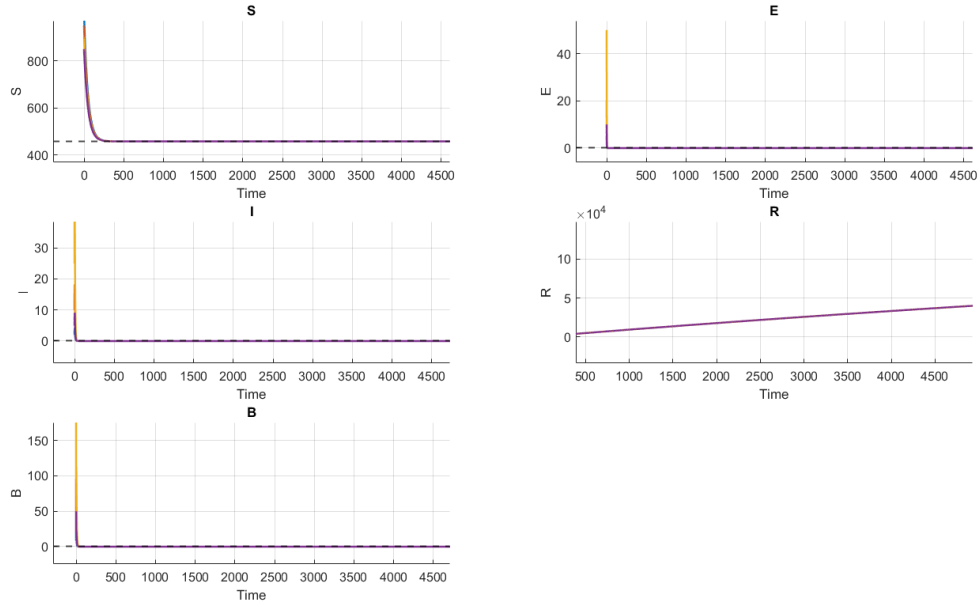


Figure 4: Stability of the disease-free equilibrium under different initial conditions.

Figure 4 shows that under the disease-free case, the numerical trajectories approach the disease-free equilibrium. In particular, the exposed, infectious, and bacterial compartments approach zero, confirming the stability prediction when $R_0 < 1$.

Figure 4: Stability of EE, $R_0 = 1.922 > 1$

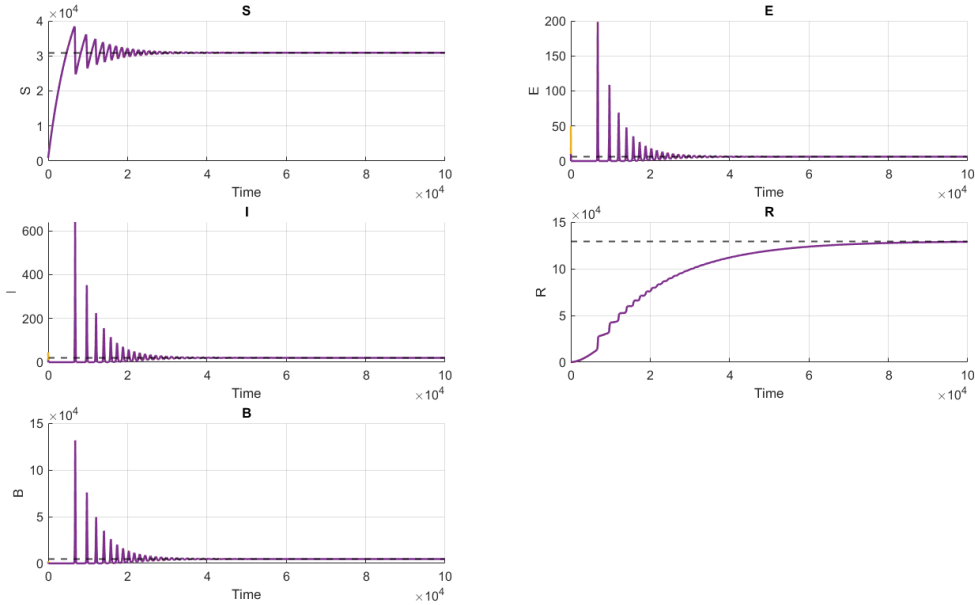


Figure 5: Stability of the endemic equilibrium under different initial conditions.

Figure 5 shows that under the endemic case, the numerical trajectories move toward the endemic equilibrium. This supports the analytical result that the endemic equilibrium is stable under the stated Routh–Hurwitz conditions when $R_0 > 1$.

4.1 Validation of Proofs

To validate the analytical results, the numerical simulations were compared with the theoretical equilibrium values derived in the mathematical analysis. The MATLAB simulations show the time-series behavior of the model compartments, while the Shiny app implementation was used as an additional verification tool by comparing the theoretical equilibrium values with the final numerical simulation values.

For the disease-free case, the parameters were chosen so that $R_0 \approx 0.0146 < 1$. Based on the stability analysis, the disease-free equilibrium is expected to be locally asymptotically stable. The simulation results agree with this result because the exposed class $E(t)$, infectious class $I(t)$, and environmental bacteria $B(t)$ approach zero. The comparison from the Shiny app implementation is shown in Table 4.

Variable	Theoretical DFE	Final Simulation	Difference
S	458.4348	458.4348	0.0000
E	0.0000	0.0000	0.0000
I	0.0000	0.0000	0.0000
R	166703.5652	156081.5895	−10621.9757
B	0.0000	0.0000	0.0000

Table 4: Comparison of theoretical disease-free equilibrium values and final Shiny app simulation values at $t = 50000$ days.

For the endemic case, the parameters were chosen so that $R_0 \approx 1.92 > 1$. Based on the analysis, the disease-free equilibrium becomes unstable and the endemic equilibrium exists biologically. The simulation results agree with this prediction because the exposed, infectious, and bacterial compartments do not approach zero. Instead, they approach positive values, showing disease persistence. The comparison from the Shiny app implementation is shown in Table 5.

Variable	Theoretical EE	Final Simulation	Difference
S	30904.5394	30869.5540	−34.9854
E	6.1646	6.0448	−0.1198
I	20.0103	19.6375	−0.3728
R	128954.8117	118994.9301	−9959.8816
B	4653.5590	4586.9316	−66.6274

Table 5: Comparison of theoretical endemic equilibrium values and final Shiny app simulation values at $t = 50000$ days.

The results show that the final simulation values are consistent with the theoretical equilibrium values. The disease-related variables $E(t)$, $I(t)$, and $B(t)$ closely follow the predicted limits in both the disease-free and endemic cases. Small differences remain because the simulation was evaluated at a finite final time rather than as $t \rightarrow \infty$.

The recovered compartment $R(t)$ converges more slowly than the other compartments because the natural death rate $\mu = 5.48 \times 10^{-5}$ is very small. Since $R(t)$ contains the term $-\mu R$, its adjustment occurs on a slow demographic time scale. This explains why the final value of $R(t)$ still has a larger difference from its theoretical equilibrium value at $t = 50000$ days. Increasing the final simulation time would allow $R(t)$ to move closer to its equilibrium value.

Therefore, the MATLAB time-series plots and the Shiny app equilibrium comparisons support the analytical stability results. When $R_0 < 1$, the disease-related compartments approach zero and the disease-free equilibrium is stable. When $R_0 > 1$, the disease persists and the solution approaches the endemic equilibrium.

5 Conclusion and Recommendations

This study formulated and analyzed a modified cholera transmission model incorporating dual transmission pathways, vaccination, and environmental disinfection. The basis model of Onuorah et al. [6] was modified by introducing an exposed compartment to represent the incubation period of cholera before individuals become infectious. This modification allowed the model to capture the biological delay between exposure and active infection, making the disease progression more realistic.

The model was shown to be mathematically well-posed in the nonnegative feasible region. This means that the solutions remain nonnegative and bounded, which ensures that the model is biologically meaningful. The disease-free equilibrium and endemic equilibrium were derived, and the basic reproduction number R_0 was obtained using the next generation matrix method. The analysis showed that the disease-free equilibrium is locally asymptotically stable when $R_0 < 1$, while it becomes unstable when $R_0 > 1$. The endemic equilibrium exists biologically when $R_0 > 1$, indicating that cholera persists in the population under sufficient transmission intensity.

The MATLAB simulations supported the analytical findings. When stronger vaccination and environmental disinfection were applied, the computed value of R_0 was less than one, and the exposed, infectious, and bacterial compartments approached zero. This confirms disease elimination under the disease-free case. On the other hand, when vaccination and disinfection were weakened, R_0 became greater than one, and the disease-related compartments persisted, confirming endemic behavior.

An important result of the modified model is the change in the observed effectiveness of interventions. In the basis model of Onuorah et al. [6], route control and environmental-related control were emphasized as important intervention mechanisms. However, in the modified model and chosen parameter setting, vaccination showed the strongest visible reduction in R_0 , while environmental disinfection still played an important supporting role by reducing the bacterial concentration in the environment. This difference suggests that adding the exposed compartment and adjusting the model structure can affect how intervention strategies influence disease dynamics.

Based on these results, a combined public health strategy is recommended. Vaccination should be strengthened to reduce the susceptible population, while environmental disinfection should be maintained to decrease bacterial concentration in contaminated water or surroundings. Reducing direct household contact with infectious individuals, improving sanitation, promoting clean water access, and strengthening early treatment are also important measures for lowering transmission. These interventions should be applied together because cholera spreads through both human-to-human and environmental pathways.

For future work, the model may be improved by fitting the parameters to real cholera outbreak data. The model may also be extended by including seasonal variation, waning immunity,

spatial movement, water sanitation levels, and time-dependent control strategies. These additions may provide a more detailed understanding of cholera transmission and improve the reliability of intervention recommendations.

AI Declaration

5.1 Scope of AI Assistance

ChatGPT and Gemini were utilized as supplementary tools in the development of this paper. These tools were used for text proofreading, grammar refinement, organization of manuscript sections, improvement of sentence clarity, and structural brainstorming. They were also used to assist in the drafting, structuring, and debugging of MATLAB simulation codes, the formatting of simulation plots, and the implementation and formatting of the Shiny app.

5.2 Academic Integrity

The authors declare that AI assistance was used only as a support tool and did not replace the group's academic work. The core mathematical derivations, analytical proofs, biological interpretations, numerical analysis, and final manuscript conclusions remain entirely the original intellectual work of the group members. All model formulation, assumptions, parameter choices, computations, simulations, interpretations, and final decisions were reviewed, validated, and approved by the authors. The authors remain fully responsible for the accuracy, integrity, and originality of the work presented in this paper.

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